Derivative Products and Their Pricing Equity Module

Top 12 Equity products Advanced Quant Guide

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December 1, 2025

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Why Banks and Hedge Funds Use Volatility Surfaces

Consistent Pricing: Market makers price 10,000+ options per underlying using a single calibrated surface. A 1% surface shift reprices entire book instantaneously, enabling automated market making.

Skew Trading: Hedge funds trade RR to bet on crash risk. If 25 put trades 6 vol points rich vs. call (RR = -6), but realized skew is only -2, selling puts/buying calls yields $2M vega P&L on $50M position.

Tail Risk Hedging: Buying BF (wing options) provides crash protection cheaply. During COVID-19, BF buyers made 15:1 returns as OTM vol spiked from 25% to 80%.

Dispersion Trading: Trading index vol surface vs. single-stock surfaces exploits correlation. If index BF = 2% vs. average single-stock BF = 4%, sell index BF, buy single-name BFprofit as correlation breaks down.

Detailed Cashflow Mechanics

For a volatility surface trade on SPY with $100M vega notional:

Skew Trade (Risk Reversal):

- Sell $50M vega of 30-day 25 puts at 22% vol, receive $2.3M premium
- Buy $50M vega of 30-day 25 calls at 18% vol, pay $1.9M premium
- Net premium: +$400k (positive carry)
- PL depends on RR change: If RR moves from -4 to -2, PL = $50M Œ 2 vol points Œ vega = +$1M

Curvature Trade (Butterfly):

- Buy $50M vega of 30-day 25 puts at 22% and 25 calls at 18%
- Sell $100M vega of 30-day ATM at 20%
- Net convexity position: Long wings, short body
- PL if BF widens from 0% to 1%: +$500k

Cashflow equation:

$$\mathrm{PL}_{\mathrm{surface}} = \sum_i \mathrm{Vega}_i \times \Delta\sigma_i + \frac{1}{2}\sum_i \mathrm{Volga}_i \times (\Delta\sigma_i)^2$$

Pricing Equations (Detailed)

Surface Construction from Market Quotes:

Given market quotes for ATM, 25 RR, 25 BF:

$$\sigma_{25\Delta\mathrm{Put}} = \sigma_{\mathrm{ATM}} + \mathrm{BF} - 0.5 \times \mathrm{RR}$$
$$\sigma_{25\Delta\mathrm{Call}} = \sigma_{\mathrm{ATM}} + \mathrm{BF} + 0.5 \times \mathrm{RR}$$

SVI Parameterization:

Stochastic Volatility Inspired model:

$$\sigma^2(k) = a + b\left(\rho(k-m) + \sqrt{(k-m)^2 + \sigma^2}\right)$$

Where $k = \ln(K/F)$ is log-moneyness. Calibrate a, b, ρ, m, σ to match market quotes.

Arbitrage-Free Constraints:

- Butterfly arbitrage: $\frac{\partial^2\sigma}{\partial K^2} \geq -\frac{2}{K}\frac{\partial\sigma}{\partial K}$

```python

    def _hagan_vol(self, K, F, T, alpha, rho, nu):
        """Hagan 2002 Approximation Formula"""
        # Handle ATM case to avoid division by zero
        if abs(K - F) < 1e-5:
            term1 = alpha / (F**(1 - self.beta))
            term2 = 1 + ( ((1-self.beta)**2 * alpha**2) / (24 * F**(2 - 2*self.beta)) +
                               (rho * self.beta * nu * alpha) / (4 * F**(1 - self.beta)) +
                               ((2 - 3*rho**2) * nu**2) / 24 ) * T
            return term1 * term2

        # Standard OTM case
        log_FK = np.log(F / K)
        FK_beta = (F * K)**((1 - self.beta) / 2)
        z = (nu / alpha) * FK_beta * log_FK

        # Chi function
        chi = np.log((np.sqrt(1 - 2*rho*z + z**2) + z - rho) / (1 - rho))

        A = alpha / (FK_beta * (1 + ((1 - self.beta)**2 / 24) * log_FK**2))
        B = z / chi
        C = 1 + ( ((1 - self.beta)**2 * alpha**2) / (24 * (F * K)**(1 - self.beta)) +
                  (rho * self.beta * nu * alpha) / (4 * FK_beta) +
                  ((2 - 3*rho**2) * nu**2) / 24 ) * T

        return A * B * C

    def calibrate(self, strikes, market_vols, F, T):
        """
        Minimize Sum of Squared Errors between Model and Market
        strikes: list of K
        market_vols: list of implied volatilities (e.g. 0.20)
        """
        def objective(params):
            alpha, rho, nu = params
            error = 0.0
            for K, vol_mkt in zip(strikes, market_vols):
                vol_model = self._hagan_vol(K, F, T, alpha, rho, nu)
                error += (vol_model - vol_mkt)**2
            return error

        # Constraints: rho in [-1, 1], alpha > 0, nu > 0
        bounds = [(0.001, None), (-0.999, 0.999), (0.001, None)]

        # Initial guess: Alpha approx ATM vol, Rho=0, Nu=0.5
        initial_guess = [market_vols[len(market_vols)//2], 0.0, 0.5]

        result = minimize(objective, initial_guess, bounds=bounds, method='L-BFGS-B')
        self.params = result.x
        return self.params

if __name__ == "__main__":
    # Example Market Data (SPX-like)
    F = 4500
    T = 1.0
    strikes = [4000, 4250, 4500, 4750, 5000]
    mkt_vols = [0.22, 0.18, 0.15, 0.13, 0.12] # Skewed surface

    sabr = SABRModel(beta=1.0) # Log-normal for equities
```

```python
        """Calculate implied convexity premium"""
        T = self.vol_swap.time_to_maturity()
        var_var = self.vol_of_vol**2 * T

        # Theoretical vol swap strike
        theor_vol_strike = np.sqrt(self.var_strike) - \
                           var_var / (8 * self.var_strike**(3/2))

        # Market vol swap strike
        market_vol_strike = self.vol_swap.strike

        # Premium = market - theoretical
        return (market_vol_strike - theor_vol_strike)

# Example: SPY vol swap
if __name__ == "__main__":
    today = dt.datetime(2025, 11, 22)
    maturity = today + dt.timedelta(days=365)

    vol_swap = VolatilitySwap(
        underlying='SPY',
        vega_notional=100_000,
        strike_vol=0.191,  # 19.1% (below var)
        start_date=today,
        maturity=maturity,
        is_long=True
    )

    # Convexity adjustment
    conv_adj = vol_swap.convexity_adjustment(
        var_swap_strike=400,
        vol_of_vol=0.30
    )

    print("\n=== Volatility Swap ===")
    print(f"Theoretical Vol Strike: {conv_adj:.1%}")
    print(f"Market Vol Strike: {vol_swap.strike:.1%}")
    print(f"Convexity Premium: {vol_swap.strike - conv_adj:.2%}")

    # MTM
    mtm = vol_swap.mark_to_market(
        realized_vol_sofar=0.17,
        current_fwd_vol=0.19,
        as_of=today + dt.timedelta(days=180)
    )

    print(f"\nMTM (6 months): ${mtm['MTM']:,.0f}")
    print(f"Vega: ${mtm['vega']:,.0f} per vol point")
    print(f"Volga: {mtm['volga']} (zero convexity)")
```

Interview Questions and Answers

Q1: Explain why volatility swaps trade below variance swap strikes.

A: Due to Jensen's inequality: $\mathbb{E}[\sigma] < \sqrt{\mathbb{E}[\sigma^2]}$. The convexity adjustment depends on vol-of-vol:

$$K_{\text{vol}} = \sqrt{K_{\text{var}}} - \frac{\text{Var}(\sigma^2)}{8K_{\text{var}}^{3/2}}$$